Methods Appendix:
Estimating Local LGBTQ Populations

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Overview

I estimate the expected LGBTQ population of every U.S. Census tract in five conceptual stages: (1) measure LGBTQ identification and gender-identity composition in Household Pulse Survey (HPS) microdata, softly calibrated to external Gallup benchmarks; (2) stabilize these rates against sampling noise using two complementary smoothing techniques — one across nearby ages, one across states; (3) project the resulting rates onto Census tracts using ACS 5-year age–sex population counts; (4) reweight the within-state spatial distribution using the observed geographic concentration of same-sex couples and singles within an age-sex cell as a proxy for the broader LGBTQ population, smoothed across space so the result doesn't show artificial jumps at PUMA boundaries; and (5) enforce a capacity constraint so no tract's estimate can exceed its underlying population. Each stage is detailed in its own section below, with the precise notation defined immediately below.

Notation

Throughout, i indexes survey respondents and w_i denotes HPS survey weights. I fix notation of gender and sex up front:

- **Sex assigned at birth**, $s \in \{M, F\}$. Since the ACS does not ask about gender identity, I begin with sex assigned at birth. Table B01001 provides tract-level counts of age-sex cells.
- **Gender identity**, $g \in \{\text{cis}, \text{nb}, \text{trans}\}$, estimated from HPS in Section 4. This is *not* the same as the final output gender bucket used for population accounting on the map. Instead, the gender identity variable simply flags an individual's relationship to their sex assigned at birth. For example, transgender respondent's gender identity is not their sex assigned at birth, so "trans" respondents with $s = M$ are trans women and those with $s = F$ are trans men.

I therefore define a third symbol, the **gender bucket** $k \in \{m, w, \text{nb}\}$ (man, woman, non-binary), combining the sex assigned at birth and gender identity:

$$k(s,g) = \begin{cases} m & (s = M,\ g = \text{cis}) \text{ or } (s = F,\ g = \text{trans}) \\ w & (s = F,\ g = \text{cis}) \text{ or } (s = M,\ g = \text{trans}) \\ \text{nb} & g = \text{non-binary} \end{cases}$$

In words: a cisgender respondent's bucket matches their sex assigned at birth; a transgender respondent's bucket does not match their sex assigned at birth. Non-binary respondents form their own bucket regardless of s. Sections 6–8 report quantities indexed by k; Section 4 reports the intermediate quantities indexed by g that k is built from.

The remaining recurring symbols are: a for exact age or an ACS age bin; $\ell \in \{\text{LG}, \text{Bisexual}, \text{Queer}\}$ for LGBT subcategory (excluding Trans, which is cross-cut with k rather than treated as a fourth member of ℓ); p for a PUMA (Public Use Microdata Area); and b for a Census tract.

1 Measurement in HPS Microdata

Let $Y_i \in \{0, 1\}$ indicate whether individual i identifies as LGBTQ. In the HPS, I mark respondents as LGBTQ if they report a sexual orientation besides straight, a trans gender identity, or both. Note that non-binary people are not automatically LGBTQ, though an overwhelming majority report a non-straight sexuality too. Sex assigned at birth s_i and age a_i are taken directly from HPS (age top-coded at 62 to limit disclosure risk in the sparse cells of older cohorts). Survey weights w_i are retained throughout all of the estimation steps below.

2 Soft Calibration to Gallup Marginals

HPS and Gallup differ in survey mode, sampling variation, and field timing, so I don't expect them to agree exactly and I don't force them to. Instead, HPS-based probabilities are *softly* calibrated toward Gallup national benchmarks, nudged toward external validity.

Each individual begins with a baseline probability $p_{0i} = \text{clipped}(Y_i)$ (clipped away from exactly 0 or 1 so the subsequent logit transform is well-defined). A logit adjustment model is estimated:

$$\text{logit}(p_i) = \text{logit}(p_{0i}) + X_i \theta,$$

where X_i includes generation effects and generation-by-sex interactions. The parameter vector θ minimizes a weighted objective combining (i) a deviation penalty that keeps estimates close to HPS baselines, and (ii) a margin penalty that discourages deviation from Gallup targets, optimized via BFGS. In practice, this keeps the vast majority of state-level estimates within one to two percentage points of Gallup's national top-line estimate of roughly 9 percent, while still letting genuine state-level heterogeneity show through.

3 State–Sex–Age LGBT Rate Estimation

Using calibrated probabilities, sex–age–state LGBT identification rates are computed as a straightforward weighted mean:

$$\widehat{P}(Y = 1 \mid s, a, \text{state}) = \frac{\sum_i w_i \, p_i^{\text{adj}}}{\sum_i w_i},$$

with effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ tracking how much genuine information backs each cell. Weekly HPS survey waves are pooled and averaged to reduce short-run noise before this step.

4 Age-Local Dirichlet Smoothing of Gender Composition

This step estimates $P(g \mid \ell, s, a)$ – the distribution of gender-identity type $g \in \{\text{cis}, \text{nb}, \text{trans}\}$ within each LGBT subcategory, sex assigned at birth, and exact age (see *Notation* above for why g is deliberately kept distinct from the final output bucket k).

The raw cell rate is

$$\hat{p}_{g,\ell,s,a} = \frac{n_{g,\ell,s,a}}{\sum_g n_{g,\ell,s,a}}, \qquad n_{\text{eff},\ell,s,a} = \frac{(\sum w_i)^2}{\sum w_i^2},$$

where $n_{g,\ell,s,a}$ denotes survey-weighted counts. Some cells are thin; in particular, non-binary and transgender respondents at a single exact age within a single sexuality category might often yield sparse cells. I address this with two smoothing passes: first borrowing strength from *nearby ages* within a state (this section), then, in Section 5, pooling the coarser LGBT subcategory shares *nationally.*

4.1 Age-Local Prior Construction

I motivate the smoothing with a concrete pattern in the data: younger AFAB (assigned female at birth) lesbians are roughly 15 percentage points more likely to identify as non-binary than AMAB gay men in the same age cohort. I also don't want a single noisy exact-age cell to dominate the estimate. Age-local smoothing does exactly this: it borrows information from *nearby* ages, weighted by how much data backs each neighbor and how close in age they are.

For a focal age a, neighboring ages $a' \neq a$ receive weights

$$w_{a'}^{(a)} = n_{\text{eff},\ell,s,a'} \, \exp\left(-\frac{|a - a'|}{\tau} \right),$$

where $\tau > 0$ controls how quickly influence decays with age distance. The age-local prior mean is then the weighted average of neighboring rates:

$$m_g^{(a)} = \frac{\sum_{a' \neq a} w_{a'}^{(a)} \, \hat{p}_{g,\ell,s,a'}}{\sum_{a' \neq a} w_{a'}^{(a)}}.$$

4.2 Dirichlet Moment Matching

A Dirichlet prior is centered at $\mathbf{m}^{(a)}$, with concentration parameter estimated via method of moments:

$$\alpha_0 = \text{mean}_g \left(\frac{m_g(1 - m_g)}{v_g} - 1 \right),$$

where v_g is the weighted variance of $\hat{p}_{g,\ell,s,a'}$ across neighboring ages a' – intuitively, how much the neighbors agree with each other. The prior vector is $\alpha_g = \alpha_0\, m_g$.

4.3 Posterior Shrinkage

The smoothed estimate for focal age a blends the raw cell rate with the age-local prior, weighted by how much real data each side has:

$$\tilde{p}_{g,\ell,s,a} = \frac{n_{\text{eff},\ell,s,a}\, \hat{p}_{g,\ell,s,a} + \alpha_g}{n_{\text{eff},\ell,s,a} + \alpha_0}.$$

A well-sampled cell (n_{eff} large relative to α_0) stays close to its raw rate; a thin cell shrinks toward the age-local prior.

5 National LGBT Subcategory Composition

For each sex s and age bin a, I also need the split between LG, Bisexual, and Queer identification after conditioning on LGBTQ identity. I pool nationally instead of estimating by state:

$$\tilde{\pi}_{\ell,s,a} = \frac{\hat{\pi}_{\ell,s,a}}{\sum_{\ell'} \hat{\pi}_{\ell',s,a}}, \qquad \hat{\pi}_{\ell,s,a} = \text{mean over HPS weeks of the national } (s,a) \text{ share identifying as } \ell.$$

Every state now applies the same LG/Bisexual/Queer mix to its own state-level, Gallup-calibrated LGBT identification rate $\widehat{P}(Y = 1 \mid s, a, \text{state})$ from Section 3 – states differ in *how many* people identify as LGBT, but not in *the mix of labels* people use.

6 ACS Projection to Census Tracts

Let $N_b^{(s,a)}$ denote the ACS 5-year population count for sex assigned at birth $s \in \{M, F\}$ and age bin a in Census tract b. I now combine every rate estimated above – identification, subcategory share, and gender-identity composition – with local ACS demographics to get an expected count for each tract, LGBT subcategory, and output gender bucket $k \in \{m, w, \text{nb}\}$:

$$\hat{E}[\text{count}_{k,\ell,b}] = \sum_{s,a} N_b^{(s,a)} \cdot \widehat{P}(Y = 1 \mid s, a, \text{state}) \cdot \tilde{\pi}_{\ell,s,a} \cdot \sum_{g\,:\,k(s,g)=k} \tilde{p}_{g,\ell,s,a}.$$

The innermost sum collapses the gender-identity-type rate $\tilde{p}_{g,\ell,s,a}$ from Section 4 into the output bucket k using the $(s, g) \mapsto k$ mapping defined in *Notation*. Because identification, subcategory, and gender-identity rates all vary with age and sex, and both vary locally, expected LGBTQ counts vary across tracts even within a single state – this section captures only the *demographic composition* channel of that variation; the next section adds a *geographic clustering* channel on top of it. For the PUMA reweighting step below, the per-cell summand of this sum – indexed by (s, g, ℓ, a, b), before summing over s, a and collapsing g into k – is what actually gets reweighted; this formula shows the result after that collapse.

Tracts that span PUMA boundaries receive population-weighted contributions from each overlapping PUMA, proportional to the area of intersection.

7 PUMA Reweighting

Spatial sorting shapes the spatial distribution of LGBTQ people across the United States. LGBTQ people cluster for many reasons, including proximity to community infrastructure, local social tolerance, idiosyncratic amenity preferences, and more. Yet the projection in Section 6 allocates LGBTQ population using only local age and sex composition – it offers no mechanism for the spatial sort of LGBTQ people beyond age and sex. I use the observed concentration of same-sex couples as a proxy for the broader LGBTQ population's spatial distribution.

Same-sex couples are an imperfect proxy, skewing older, wealthier, better-educated, and more likely to identify as gay or lesbian than the broader LGBTQ population. Within metropolitan areas, their spatial Concentration might systematically differ other LGBTQ sub-groups, particularly young, non-binary and transgender populations. Rather than apply a couples-based tilt to the whole LGBTQ population, I blend it with a second spatial proxy: the geographic concentration of *singles* of the same sex-age cell. The weight placed on couples relative to single depends on the national age-sex cell marriage rate in the HPS.

7.1 Coupled and Singles Spatial Signals

I construct a same-sex-couple signal from ACS Table B09019 counts of same-sex (SS) and opposite-sex (OS) couples by PUMA, unchanged from before: PUMAs with fewer than 100 total couples are dropped as too noisy to trust, and the raw same-sex share is winsorized to the 1st–99th percentile across PUMAs before transforming, to keep a handful of tiny, extreme PUMAs from dominating the calibration,

$$\hat{q}_p = \text{winsorize}_{[0.01,\,0.99]}\left(\frac{\text{SS couples}_p}{\text{SS couples}_p + \text{OS couples}_p}\right).$$

I normalized to a spatial share:

$$\rho_p^{\text{couple}} = \frac{\text{SS couples}_p}{\sum_{p'} \text{SS couples}_{p'}}.$$

The singles signal comes from ACS Table B12002 (sex by marital status by age), for birth sex s and age bin a:

$$\rho_p^{\text{single}}(s, a) = \frac{\text{Not-married}_{p,s,a}}{\sum_{p'} \text{Not-married}_{p',s,a}},$$

where "not married" includes everyone except people currently married with a spouse present.

PUMAs with fewer than 100 such people in a given (s, a) are dropped, the same threshold applied to the couples signal.

7.2 Coupled-Share Blend Weight

From the HPS, I estimate the national married share of people in cell (s, g, ℓ, a). HPS lacks an option for unmarried cohabiting partners question, so "married" is an imperfect proxy for "coupled", understating true coupled cohabitation particularly among younger people. I smooth the married share across nearby ages exactly as in Section 4 (the same age-local, effective-sample-size-weighted decay, but for a single Beta-distributed share rather than a Dirichlet-distributed composition vector), then winsorized to $[0.05, 0.95]$ so no cell relies entirely on one spatial signal:

$$\hat{\mu}_{s,g,\ell,a} = \text{winsorize}_{[0.05,\,0.95]}\big(\text{smoothed married share of cell } (s, g, \ell, a)\big).$$

7.3 PUMA Calibration Factors

For each PUMA p and cell (s, g, ℓ, a), define the uncalibrated LGBTQ share of the state total projected in Section 6, now computed *within* the cell rather than pooled across all cells:

$$S_p(s, g, \ell, a) = \frac{\sum_{b \in p} \hat{E}[\text{count}_{s,g,\ell,a,b}]}{\sum_{p'} \sum_{b \in p'} \hat{E}[\text{count}_{s,g,\ell,a,b}]}.$$

The blended spatial share for cell (s, g, ℓ, a) is a weighted average of the two signals above, using that cell's own coupled share as the weight:

$$\rho_p(s, g, \ell, a) = \hat{\mu}_{s,g,\ell,a}\, \rho_p^{\text{couple}} \;+\; (1 - \hat{\mu}_{s,g,\ell,a})\, \rho_p^{\text{single}}(s, a).$$

From here the tilt machinery is unchanged from the original single-signal version, just re-indexed by cell. The blended signal is logit-transformed and mean-centered relative to the cell's own uncalibrated distribution,

$$\tilde{\sigma}_p(s, g, \ell, a) = \text{logit}(\rho_p(s, g, \ell, a)) \;-\; \sum_{p'} S_{p'}(s, g, \ell, a)\, \text{logit}(\rho_{p'}(s, g, \ell, a)),$$

and the tilt factor, calibrated target share, and calibration factor follow exactly as before:

$$T_p(s, g, \ell, a) = \exp\big(\gamma\, \tilde{\sigma}_p(s, g, \ell, a)\big),$$

$$S_p^*(s, g, \ell, a) = \frac{S_p(s, g, \ell, a)\, T_p(s, g, \ell, a)}{\sum_{p'} S_{p'}(s, g, \ell, a)\, T_{p'}(s, g, \ell, a)},$$

$$\hat{c}_p(s, g, \ell, a) = \frac{S_p^*(s, g, \ell, a)}{S_p(s, g, \ell, a)}.$$

γ keeps its original role as a single global conservative dial (site still defaults to $\gamma = 0.5$): it governs how strongly *any* cell's reweighting responds to its blended spatial signal, while the coupled-share weight $\hat{\mu}$ is what determines *which* spatial signal – couples or singles – that cell is actually being tilted toward.

7.4 Inverse-Distance-Weighted Smoothing to Tract Level

Applying $\hat{c}_p(s, g, \ell, a)$ uniformly to every tract within PUMA p produces a step-function surface, with hard discontinuities at PUMA boundaries. Yet those jumps stem from where the Census Bureau happened to draw PUMA lines, not real demographic transitions. Hence, I smooth the calibration factors to the tract level using inverse-distance weighting (IDW) instead of applying them as a step function.

Let $\mathbf{x}_b$ and $\mathbf{x}_p$ denote the centroids of tract b and PUMA p. The IDW-smoothed calibration factor for tract b and cell (s, g, ℓ, a) is

$$\tilde{c}_b(s, g, \ell, a) = \frac{\sum_p \|\mathbf{x}_b - \mathbf{x}_p\|^{-2}\, \hat{c}_p(s, g, \ell, a)}{\sum_p \|\mathbf{x}_b - \mathbf{x}_p\|^{-2}}.$$

A tract deep inside a single PUMA receives a factor close to that PUMA's $\hat{c}_p$; a tract near a PUMA boundary receives a blend of the neighboring PUMAs' factors, weighted by inverse squared distance. IDW smoothing preserves geographic heterogeneity while producing a spatially continuous surface with no visible seams.

All subcategory estimates in tract b and cell (s, g, ℓ, a) are multiplied by $\tilde{c}_b(s, g, \ell, a)$ and then summed over s, a and collapsed over g into the output bucket k, exactly as in Section 6; the whole state is then rescaled so totals match the uncalibrated projection:

$$\text{scale} = \frac{\sum_b \hat{E}[\text{LGBTQ}_b]}{\sum_b \tilde{c}_b\, \hat{E}[\text{LGBTQ}_b]}.$$

8 Capacity Constraint

The PUMA reweighting above is a multiplicative tilt, which lacks a natural ceiling. In high-concentration areas, reweighting and state-level rescaling can push a tract's estimated count for an output gender bucket k above that bucket's actual population. Yet no tract can contain more LGBTQ men than it contains men (or women or envies).

For output gender bucket $k \in \{m, w, \text{nb}\}$ and tract b, let $\hat{N}_{k,b}^{\text{LGBTQ}}$ denote the post-calibration LGBTQ count and $N_{k,b}$ the bucket's total population. I cap each bucket at its population ceiling and scale every LGBTQ subcategory within that bucket down proportionally, preserving relative composition:

$$\kappa_{k,b} = \min\left(1,\ \frac{N_{k,b}}{\hat{N}_{k,b}^{\text{LGBTQ}}}\right),$$

with every subcategory count for bucket k in tract b multiplied by $\kappa_{k,b}$. Imposition of the capacity constraint is rare.